

# TRAFFIC VIOLATION & FINE MANAGEMENT SYSTEM

## Abstract

The Traffic Violation & Fine Management System is a menu-driven console-based application developed using Core Java, JDBC, and MySQL. The system helps in recording traffic violations, calculating fines, tracking repeat offenders, and managing payment status efficiently.

## Objectives

- To record and manage traffic violation details.
- To calculate fines based on violation type and repeat offenses.
- To store and retrieve violation records using a MySQL database.
- To follow layered architecture for clean and maintainable code.

## Technologies Used

| Technology  | Description                         |
|-------------|-------------------------------------|
| Java        | Core Java for application logic     |
| JDBC        | Database connectivity               |
| MySQL       | Backend database                    |
| Eclipse IDE | Development environment             |
| GitHub      | Version control and project hosting |

## System Architecture

The project follows a layered architecture consisting of DTO, DAO, Service, and Main layers. This separation ensures better readability, maintainability, and scalability of the application.

## Modules Description

**DTO (Data Transfer Object):** Stores traffic violation data without business logic.

**DAO (Data Access Object):** Handles all database operations using JDBC.

**Service Layer:** Contains business logic such as fine calculation and repeat offender handling.

**Main Module:** Provides a menu-driven interface for user interaction.

## Key Features

- Add traffic violation details.
- Automatic fine calculation.
- Detection of repeat offenders.
- View and manage violation records.
- Menu-driven user interface.

## Conclusion

The Traffic Violation & Fine Management System provides an efficient solution for managing traffic violations using Java and database technologies. It demonstrates practical implementation of JDBC, object-oriented programming, and layered architecture suitable for academic and real-world applications.